MIL-P-20442B (MU)
8 May 1968
SUPERSEDING
MIL-P-20442A
30 January 1964

MILITARY SPECIFICATION

PLUGS, LIFTING (EYEBOLT TYPE) FOR PROJECTILES

1. SCOPE

1.1 This specification covers three types of lifting plugs.

1.2 Classification.-The lifting plugs shall be of the following types as specified (see 6.1).

a. Type C
b. Type G
c. Detector Type

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-A-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1320

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DRAWINGS

ARMY

75-14-42 - Plugs, Lifting (Eyebolt Type) for Projectile
10520074 - Plug, Lifting Type G
10520911 - Plug, Lifting, Type-Detonator Assembly

PUBLICATIONS

ARMY

ET-6112851 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Plug.-The plug shall comply with all requirements specified on Drawing (dwg.) 75-14-42, 10520074 or 10520911 as applicable, and with all requirements in applicable specifications.

3.2 Tension.-The plug shall sustain load of 8,000 pounds without any sign of failure, when tested as specified in 4.4.1.

3.3 Ductility.-The plug shall withstand deformation of the eye without cracking, when tested as specified in 4.4.2.

3.4 Fracture (applicable to malleable iron only).-The plug shall be free from porosity and shrinks or discontinuous structure, when tested as specified in 4.4.3.

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3.5 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, dust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent.

3.6 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

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f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five assemblies and twenty-five each of every component which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

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requirements of the applicable drawing. All paint shall be dry

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspections provisions

4.3.1 Lot formation.-The term lot as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

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4.3.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.3.2.1 Plug, type G or C, prior to painting (see dwg. 10520074 or 75-14-42).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Diameter around eye section, minimum (min)	Gage	01001
102.	Length to shoulder, min. ...	Gage	01002
103.	Pitch diameter of thread ...	Gage	01003
104.	Major diameter of thread ...	Gage	01004
105.	Wall thickness of threaded portion, min.	Gage	01005
106.	Metal split or cracked	Visual	01006
Minor:	AQL 4.00 percent		
201.	Eye hole diameter, min.	Gage	01007
202.	Profile of eye hole, min. ..	Gage	01008
203.	Depth of core, maximum (max)	Gage	01009
204.	Length of thread undercut, max.	Gage	01010
205.	Diameter of flange	Gage	01011
206.	Thickness of flange, min. ..	Gage	01012
207.	Diameter of undercut	Gage	01013
208.	Evidence of poor workmanship (see 3.5)	Visual	01014

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4.3.2.2 Plug, type G and C (see dwg. 10520074 or 75-14-421.)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	02001
202.	Compound missing from thread.	Visual	02002

4.3.2.3 Washer (see dwg. 10520913 covering a detail of dwg. 10520911).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Outside diameter	Gage	03001
202.	Inside diameter	Gage	03002
203.	Thickness	Gage	03003

4.3.2.4 Screw, (see dwg. 10520912 covering a detail of dwg. 10520911).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pitch diameter of thread	Gage	04001
102.	Major diameter of thread	Gage	04002
103.	Thread undercut, max.	Gage	04003

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 2.50 percent		
201.	Length including threads undercut and collar	Gage	04004
202.	Thickness of collar	Gage	04005
203.	Thickness of head	Gage	04006
204.	Diameter of collar	Gage	04007
205.	Width across flats of head	Gage	04008
206.	Finish improper	Gage	04009
207.	Evidence of poor workmanship (see 3.5)	Visual	04010

4.3.2.5 Plug, lifting detector type (see dwg. 10520910 covering a detail of dwg. 10520911).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Diameter around eye section, min.	Gage	05001
102.	Length to shoulder, min.	Gage	05002
103.	Major diameter of plug threaded	Gage	05003
104.	Pitch diameter of plug threaded	Gage	05004
105.	Pitch diameter of screw thread.	Gage	05005
106.	Minor diameter of screw thread.	Gage	05006
107.	Wall thickness of threaded portion, min.	Gage	05007
108.	Concentricity of collar surface with screw thread	Gage	05008
109.	Position of screw hole	Gage	05009
110.	Metal split or cracked	Visual	05010
Minor:	AQL 4.00 percent		
201.	Eye hole diameter, min.	Gage	05011
202.	Profile of eye hole, min.	Gage	05012
203.	Depth of core, max.	Gage	05013
204.	Length of thread undercut, max.	Gage	05014
205.	Diameter of flange	Gage	05015
206.	Thickness of flange, min.	Gage	05016
207.	Diameter of undercut	Gage	05017
208.	Evidence of poor workmanship (see 3.5)	Visual	05018

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4.3.2.6 Plug lifting detector type (see dwg. 10520910 covering a detail of dwg. 10520911).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	06001
202.	Compound missing from thread.	Visual	06002

4.3.2.7 Plug lifting detector type (see dwg. 10520911).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 0.65 percent		
201.	Screw or washer missing or improperly seated	Visual	07001

4.3.3 Testing (see 3.2, 3.3 or 3.4) Major defect

4.3.3.1 Sample.-Two lifting plugs shall be randomly selected from each heat treatment batch for the tension, ductility and fracture test. If any sample fails to comply with the requirement the batch represented by samples shall be rejected.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-6112851 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Tension.-The lifting plug shall be screwed into a threaded plate or other suitable fixture and a load of 8,000 pounds applied through the medium of a 3/4 inch diameter bolt or pin passing through the eye of the plug. The full load shall be transmitted through the screw threads of the plug.

4.4.2 Ductility.-The plugs used in the tension test shall be cooled to minus 40 degrees Fahrenheit (F.) for a minimum of 30 minutes. The eye of each plug shall be flattened by either pressure or blows in any direction until the inner diameter of the eye is reduced by 0.050 inch while at minus 40 degrees F.

4.4.3 Fracture.-The body of each sample malleable iron casting shall be broken through the thickest section for inspection. The plug shall be observed for discontinuous structure porosity or shrinks.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-Packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be

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submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1320-A-142

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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